SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1703**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

*I **NEELAM NANDINI [192472277]** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N.Nandini.

Annexure A

Analytical Problem Solving

1. **Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.**

Ans: Water Jug Problem (4-gallon and 3-gallon)

Initial State: (4G,3G) = (0,0)

Step	**Action**	**State (4G,3G)**
1	Fill the 3-gallon jug	(0,3)
2	Pour 3G into 4G	(3,0)
3	Fill the 3G again	(3,3)
4	Pour from 3G to 4G until 4G is full	(4,2)
5	Empty the 4G jug	(0,2)
6	Pour remaining 2G from 3G into 4G	(2,0)

The 4-gallon jug contains exactly 2 gallons, which is the required goal.

2. **Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.**

   i. **What are the percept's for this agent?**

Percepts are the information received through sensors.

- Camera images
- Rock and soil samples
- Temperature
- Atmospheric pressure
- Terrain information
- Obstacle detection
- GPS/position data
- Battery level

   ii. **Characterize the operating environment.**

- Partially Observable
- Deterministic
- Sequential
- Dynamic
- Continuous
- Single Agent

iii. **What are the actions the agent can take?**

- Move forward/backward
- Turn left/right
- Avoid obstacles
- Capture images
- Collect rock samples
- Drill rocks
- Analyze soil
- Send data to Earth

iv. **How can one evaluate the performance of the agent?**

The Mars Rover performs well if it:

- Collects correct samples
- Avoids obstacles
- Conserves battery power
- Completes mission successfully
- Sends accurate scientific data
- Travels safely

v. **What sort of agent architecture do you think is most suitable for this agent? Why?**

Model-Based Goal-Based Agent
Reason:

- Maintains an internal model of the environment.
- Plans routes toward mission goals.
- Makes decisions even when all information is not available.
- Adapts to changes in terrain.

3. **Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.**

Ans:

Initial State

An empty 8 × 8 chessboard.

State Space

Any arrangement of queens on the chessboard.

Operators

Place one queen in a valid row of the next column.

Goal Test

All 8 queens are placed such that:

- No two queens are in the same row.
- No two queens are in the same column.
- No two queens are on the same diagonal.

Path Cost

Each move has equal cost.

Total cost = Number of queens placed.

Search Strategy

Backtracking Search (Depth First Search)

- Place one queen.
- Check whether it is safe.
- If not safe, backtrack.
- Continue until all 8 queens are safely placed.

4. **A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.**

Ans:

Goal-Based Problem-Solving Agent

Reason:

The customer's goal is to reach the destination by selecting the most suitable cab (Mini, Micro, Sedan, Prime, Shared, etc.).

Pseudocode:

START

Input Source and Destination

Display available cab types

Customer selects a cab

Calculate fare and estimated arrival time

Confirm booking

Assign nearest available driver

Driver reaches pickup location

Customer travels to destination

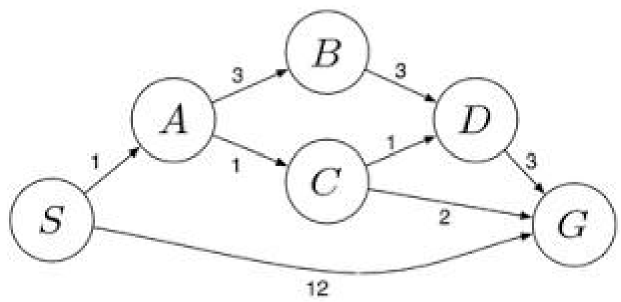
Complete trip

Generate bill

5. **A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.**

   The delivery network is represented as a weighted graph:

   Ans: Uniform Cost Search (UCS)



Algorithm

- Start from source node S.
- Insert S into the priority queue with cost 0.
- Remove the node with the lowest path cost.
- If the node is the goal G, stop.
- Otherwise, expand the node.
- Add all child nodes with cumulative costs.
- Repeat until G is reached.

Properties of UCS

- Completeness: Yes
- Optimality: Yes
- Data Structure Used: Priority Queue
- Expands the node with the lowest cumulative cost first.
- Returns the least-cost path from S to G.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem